

# Jacob Berger

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## EDUCATION

### Oregon State University Honors College

*Master of Science in Computer Science (AI Concentration)*

*B.S. in Applied Computer Science & Computational Mathematics*

Corvallis, OR

**Expected June 2027**

**Sept 2022 – June 2026**

- **Status:** Admitted to Accelerated Master's Program | **GPA:** 3.9+
- **Relevant Coursework:** Analysis of Algorithms, Machine Learning, Linear Algebra II. **Graduate:** Deep Learning, Operating Systems II, Numerical ODEs, Math of Data Science, Computational Number Theory

## RESEARCH

### Honors Thesis: Dynamic Constraints for LLMs

*Researcher (Advisor: Dr. Lizhong Chen, STAR Lab)*

Corvallis, OR

**June 2025 – Present**

- Leading an honors thesis project to develop a dynamic constraint system for large language models, improving text generation for applications in computational semantics.

### Senior Capstone Project: AI for Home Automation

*Project Team Member*

Corvallis, OR

**September 2025 – Present**

- Collaborating in a senior design team to integrate Large Language Models with the Home Assistant IoT platform, enabling users to control devices and create complex automations using natural language.

## EXPERIENCE

### NLP SaaS Startup

*Language Model and Software Engineer*

Portland, OR

**March 2023 – Present**

- Architected end-to-end data pipelines for training and fine-tuning specialized large language models.
- Led and co-authored the team's approved Phase I Small Business Innovation Research (SBIR) proposal.
- Engineered and deployed cloud infrastructure on AWS, GCP, and Vercel to train and host services.

### MU Lanes and Games

*Head Mechanic*

Corvallis, OR

**December 2022 – Present**

- Managed maintenance projects and diagnostics for complex electromechanical systems.

### MU Student Activities and Advisory Board

*Vice-Chairman*

Corvallis, OR

**June 2024 – Present**

- Oversaw budget preparation and facilitated strategic board meetings for OSU's Memorial Union.

## SKILLS

- **Languages:** Python, C++, Java, Javascript, Matlab, Bash/Shell, x86 Assembly
- **Machine Learning:** PyTorch, HuggingFace Transformers, LLM Fine-tuning (LoRA), RAG Pipelines
- **Systems & Infrastructure:** Linux (Debian), Proxmox VE, Virtualization (LXC/KVM), GPU Passthrough
- **Cloud & DevOps:** AWS (EC2/S3), Google Cloud Platform (GCP), Git, Vercel
- **Languages (Human):** Fluent in Spanish (9 years of language immersion education)

## AWARDS

- **Putnam Math Competition Award:** Top 3 Finisher at Oregon State University.
- **Finley Academic Excellence Scholarship:** \$20,000 scholarship recipient for academic merit at OSU.

## EXTRACURRICULARS

- **Competitive Math:** Competitor in Oregon State Putnam and COMAP Math Modeling competitions.
- **Competitive Sports:** Founded OSU's national collegiate bowling team (All-Star qualifier); served as Rose City Rowing Club team leader (2-time national qualifier).